Database Development and Design (SOE201)

Lab 1: Tutorial on Database Storage

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AIAT

Q1

- Given a disk with the following characteristics:
  - There are 2^{14}=16384 tracks per surfaces
  - There are 2^{7}=128 sectors per track
  - There are 2^{12}=4096 byte per sector
  - The disk rotates at 7200rpm; i.e. it makes one rotation in 8.33 milliseconds
  - To move the head arm between cylinders (tracks) take one milliseconds to start and stop, plus one additional millisecond for every 1000 cylinders travelled.
- Questions:
  - 1. what is the time to take one track movement?
  - 2. what is the time to move the head from innermost track to outmost track?

Q2

- Given a disk with the following characteristics:
  - There are 2^{14}=16384 tracks per surfaces
  - There are 2^{7}=128 sectors per track
  - There are 2^{12}=4096 byte per sector
  - The disk rotates at 7200rpm; i.e. it makes one rotation in 8.33 milliseconds
  - To move the head arm between cylinders (tracks) take one milliseconds to start and stop, plus one additional millisecond for every 1000 cylinders travelled.
- Questions:
  - 1. Assume that there is no gap between sectors and each block occupies 4 sectors. what is the minimum time to read a block ?
  - 2. What is the maximum time to read a block?
  - 3. What is the average time?

Q3

- Suppose that a relation called student holds 25,000 tuples, which are stored as fixed length and fixed format records. The length of each tuple is 350 bytes. The key attribute, student_ID, occupies 10 bytes and another attribute address occupies 50 bytes. The records are sequentially ordered by student_ID and stored in a number of blocks. Each block has the size of 4,096 bytes (i.e. 4 Kilobytes). Assume that a complete record or an index entry must be stored in one block.
  - (1) How many blocks are needed to store the relation student?
  - (2) Consider creating a primary index on the *student_ID* attribute. Each index entry contains a search key and a 10-byte long pointer to the records. Suppose the primary index is sparse (i.e. one index entry for one block), compute the number of blocks needed to store the index.

Q4

- A relational database contains the following two relations:
  - Relation *student*:
    - Tuples are stored as fixed-format, fixed-length records, each of 300 bytes.
    - Each tuple contains a key attribute *sID* of length 20 bytes; other fields and the record header make up the rest.
    - There are 1,000 tuples.
  - Relation *module*:
    - Tuples are stored as fixed-format, fixed-length records, each of 200 bytes.
    - Tuples contain attribute *module.sID* (ID of a student who takes a module), referencing the *student.sID*.
    - There are 5,000 tuples.
- Assume that each student must take exactly 5 modules. The student records are sequentially ordered by *sID* and the "clustered disk organisation" strategy is used. This means that, for each student record, the 5 module records also reside in the same block. Also, assume that a record does not span over more than one block, and the size of each block is 4,096 bytes.

Q4 – cont'd

- (1) Calculate the number of disk blocks needed to store the two relations.
- (2) Suppose that a dense primary index is to be created on *sID* for the *student* relation, i.e. one index entry for each tuple. Each index entry includes a key and a 10-byte pointer to data (an 8 byte block ID and a 2 byte offset). How many disk blocks are needed to store the index?
- (3) Suppose that a sparse primary index is to be created on *sID* for the *student* relation, i.e., one index entry for each disk block. Each index entry includes a key and a 10-byte pointer to data (an 8 byte block ID and a 2 byte offset). How many disk blocks are needed to store the index?

Q5

- A relation named product has 10,000 tuples, which are stored as fixed length and fixed format records; each has the length of 50 bytes. Tuples contain the key attribute productID with length of 2 bytes. The tuples are stored sequentially in a number of blocks, ordered by productID. Each block has the size of 4,096 bytes and each tuple is fully contained in one block.
  - (1) What would be the number of disk blocks needed to store the relation product?
  - (2) Suppose that a sparse primary index (i.e. one index entry for one block) on productID is to be created. A 10-byte pointer to the actual tuples is needed for each index entry. Each index entry is also fully contained in one block. What would be the number of disk blocks needed to store the entire index?

Q6

- 1) Relation students has 30,000 tuples, which are stored as fixed length and fixed format records. Each tuple has a length of 30 bytes. Tuples contain the key attribute studentID with a length of 2 bytes; a non-key attribute name with a length of 4 bytes. The tuples are stored sequentially in a number of blocks, ordered by name. Assume that each block has a size of 4,096 bytes and each tuple is fully contained in one block. What would be the number of disk blocks needed to store the relation students?
- 2) Based on the same information as in Question 1), suppose that a dense primary index is to be created. A 6-byte pointer to the actual tuple is needed for each index entry. Each index entry is also fully contained in one block. What would be the maximum number of blocks needed to store this index?
- 3) Based on the same information as in Question 1) and Question 2), suppose that a sparse secondary index on the studentID attribute is to be created. Which of the two indices, the dense primary index and sparse secondary index, is more efficient in looking-up tuples with given search keys? Justify your answer.